# Md Shamiul Islam Riyad

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## SUMMARY

CSE undergraduate at United International University (graduating 2027). Most of my project work is backend development in C# and ASP.NET Core with PostgreSQL: REST APIs, JWT login with role checks, EF Core data models, and order and account logic. I have also built the React frontends that call these APIs.

## TECHNICAL SKILLS

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| **Languages** | C#, SQL, JavaScript, C++, Java |
| **Backend** | ASP.NET Core Web API (.NET 8 and 10), Entity Framework Core, REST APIs, FluentValidation, JWT authentication, role-based authorization, SignalR, background services |
| **Databases** | PostgreSQL, relational schema design, EF Core code-first migrations, SQL queries |
| **Testing and tools** | xUnit with EF Core InMemory, Swagger and .http request files for manual API testing, Git and GitHub |
| **Frontend** | React, Axios (used to connect my APIs to a UI) |

## PROJECTS

**ShilpoHubBD** – marketplace and business platform for Bangladeshi artisans github.com/shamiulriyad/ShilpoHubBD

*ASP.NET Core 8, EF Core, PostgreSQL, JWT, FluentValidation, SignalR, React. Team of 5; I wrote most of the backend.*

- Wrote entities and 100+ EF Core migrations for a 312-table PostgreSQL schema, with the code split into Domain, Application, Data, Infrastructure and Api projects.
- Implemented JWT login with refresh tokens that are stored as hashes, replaced on every refresh and revoked on logout. Added 9 user roles, with ownership checks so producers can only manage their own products, orders and stock.
- Added an inventory log that records every stock change, rejects adjustments that would make stock negative, and lists low-stock products per producer. A global exception handler maps validation, not-found and conflict errors to 400, 404 and 409 responses.

**Avenzaq** – multi-vendor e-commerce backend github.com/shamiulriyad/Avenzaq

*ASP.NET Core (.NET 10), EF Core, PostgreSQL, JWT, SignalR, React*

- Wrote checkout as one database transaction: the cart is split into a sub-order per seller with that seller's commission and net amount, stock is reduced per product or variant, and the whole order rolls back if any item is out of stock.
- Protected Customer, Seller and Admin endpoints with JWT role checks. A seller needs admin approval before managing orders and can only change the status of their own sub-orders.
- Saved order notifications (placed, confirmed, shipped, delivered) in PostgreSQL with paging and unread counts, and pushed them to each user through a SignalR hub.

**E-Learning Platform API** – university course and competition platform github.com/shamiulriyad/DatabaseHub

*ASP.NET Core 8, EF Core, PostgreSQL, JWT, SignalR, React. Team project; I built the backend.*

- Designed a 48-table PostgreSQL schema (universities, departments, courses, enrollments, assignments, competitions) with EF Core code-first, using foreign keys, unique indexes and delete rules, and changed it through migrations.
- Built 24 REST controllers with JWT authentication, Student, Teacher and Admin role checks, and BCrypt password hashing.
- Wrote a background service that moves competitions from Upcoming to Ongoing to Completed on schedule, and used SignalR to push course and community updates.

## EDUCATION

**BSc in Computer Science and Engineering**, United International University (UIU), Dhaka 2023 – 2027 (expected)

## ACHIEVEMENTS

- 2nd runner-up, UIU CSE Project Show (March 2026), with the E-Learning platform above.
- Took part in 7 hackathons in 2026. Our team placed in the top 70 of 700+ teams at Infinity AI BuildFest.
- Solved 575+ problems on Codeforces, CodeChef and LeetCode.